## 1 Asymmetric KV budget

Split the index-storage budget

$$B_{KV} = B_K + B_V$$

K: 6 bits (avg.) | V: 2

8 index bits per K/V pair on average

**Keys (K)**

Attention weights

$T \times d_{head}$

**K4 / K8**

6 bits on average

**Values (V)**

Weighted sum

$T \times d_{head}$

**V2**

2 bits per value

One matrix per KV head

## 2 Boundary allocation

28-layer Qwen3-1.7B

**Layers 1–7**
Self-attention | FFN
Values: V2 — K8

**Layers 8–21**
Self-attention | FFN
Values: V2 — K4

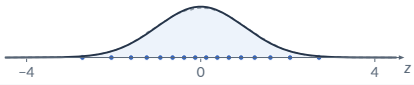
**Layers 22–28**
Self-attention | FFN
Values: V2 — K8

K8: 14 layers · K4: 14 layers
V2 in every layer

## 3 Bit-width-aware quantizer selection

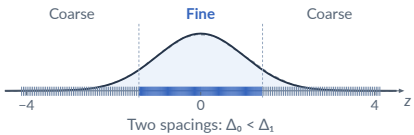
Gaussian model · KV samples · Reconstruction levels

**K4 · Lloyd–Max** — 16 levels




Threshold indexing + level lookup

**K8 · PU-K8 (ours)** — 256 levels



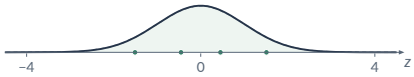

Two spacings: $\Delta_0 < \Delta_1$

Arithmetic indexing + HVX decoding

$\mathcal{C}^+ = a_r + k\Delta_r$

Arithmetic reconstruction

**V2 · Lloyd–Max** — 4 levels




Threshold indexing + level lookup